

Evaluation of Efficient Electrocardiomatrix-based Identification Using Deep Learning Methods

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Abstract—Nowadays, biometric systems are essential for identification and authentication. This study proposes deep learning models for user identification using electrocardiomatrices (ECMs) as input. The proposed models are evaluated from different perspectives to assess their efficiency and performance for user identification. One aspect involves comparing convolutional and recurrent neural models. In contrast, another element consists of testing the models with different numbers of ECMs and beats per frame for generating the ECMs. The results are based on an analysis of three different electrocardiogram databases.

Index Terms—User identification, Biometric systems, ECG, ECM, Deep learning

I. INTRODUCTION

Biometric screening involves the statistical analysis of biological data, which is crucial in identifying individuals. This analysis utilizes various signals such as fingerprints, voice recognition, and human electrocardiogram (ECG) to efficiently identify individuals [1]. ECG is particularly valuable for identifying heart and cardiovascular diseases, as well as for biometric identification [2], [3]. The raw ECG signals are complex to analyze, so electrocardiomatrix (ECM) is used to simplify the representation. ECMs aid in detecting arrhythmias and diagnosing heart diseases [4] and also contribute to biometric identification [5]. A segment is divided into subsegments to generate an ECM frame, each containing a fixed number of beats. These subsegments are then arranged in a matrix and subjected to a heatmap procedure to create the ECM frame.

Fuster-Barcelo et al. proposed a CNN model for biometric identification using ECMs. They utilized a convolutional layer followed by max pooling and a dense layer to identify the corresponding individual ECMs. They experimented with 3, 5, and 7 beats per frame (bpf) for generating ECMs [5]. However, they did not test the impact of different numbers

of ECMs for each individual. This is important for efficient people identification - the fewer, the better.

This paper proposes four different deep-learning methods for identification systems that input electrocardiomatrices. The identification system aims to act as a one-against-all system, where the model can identify the corresponding users based on their ECMs. To ensure their applicability across different scenarios, these models' robustness was tested on various datasets, including MIT-BIH, NSRDB, and PTBDB. Furthermore, this paper aims to revolutionize user identification by proposing a method to identify users with the fewest possible number of ECMs, potentially reducing the burden on users and the system. Put simply, we tested different scenarios, each based on different learning models and numbers of heartbeats, to understand the impact of these variables on generating the ECM models. The latter is essential for efficient user identification. The identification models were also tested for different beats per frame rates (bpfs). The results confirm that fewer bpfs and ECMs lead to more efficient identification. In conclusion, this study's contribution is significant.

- Proposing three identification systems using CNN models.
- proposing an RNN-based learning model for system identification.
- Generating ECMs from three ECG datasets and applying learning models on ECMs.
- Testing variations in bpfs and ECMs to achieve efficient identification with the lowest number of features.
- A comprehensive results evaluation.

The structure of this study is as follows. Section II describes the datasets and introduces the proposed identification models. Section III evaluates the results. Section IV concludes the

paper.

II. MATERIALS AND METHODS

This section provides the datasets and the proposed methods. We utilized three ECG datasets, i.e., MIT-BIH, NSRDB, and PTBDB.

A. Datasets

To generate the ECMs, we utilize three different databases, i.e., the MIT-BIH Arrhythmia database, the NSRDB database, and the PTBDB database. All datasets are publicly accessible on the PhysioNet platform, as referenced in [6]. The ECG signals from these datasets can be retrieved from the .ecg files corresponding to individual users.

The MIT-BIH Arrhythmia database: consists of 48 thirty-minute segments of two-channel ambulatory ECG recordings sourced from 47 individuals under examination at the BIH Arrhythmia Laboratory. Among these, twenty-three segments were drawn randomly from a pool of 4000 twenty-four-hour ambulatory ECG recordings gathered from a diverse cohort of inpatients (approximately 60%) and outpatients (approximately 40%). The remaining 25 segments were deliberately chosen from the same collection to encompass less prevalent yet clinically significant arrhythmias, ensuring a comprehensive representation beyond a limited random selection [7].

The NSRDB database: includes 18 long-term ECG recordings from individuals who were referred to the Arrhythmia Laboratory. These recordings were obtained from subjects without notable arrhythmias, encompassing five men aged between 26 and 45 and thirteen women aged between 20 and 50 [6].

The PTBDB database: comprises 549 entries from 290 individuals, ranging in age from 17 to 87 years old, with an average age of 57.2. It includes data from 209 males with an average age of 55.5, and 81 females with an average age of 61.6; however, the ages of 1 female and 14 male participants were not recorded. Among these subjects, 52 are healthy, and 216 have various heart conditions. Clinical summaries for 22 subjects are missing. Each individual's data is captured in one to five records [8].

B. Proposed methods

The proposed methods consisted of two main steps: ECM generation and model learning.

1) *ECM generation:* To generate the ECMs, we first decompose the ECG segment into individual beats. We cleaned the signals using a bandpass filter between 5 to 15 Hz. Then, we applied Pan-Tompkins to detect the R-peaks [9]. Lastly, we normalized data using the min-max approach. Having R-peaks, we can decompose the source signal into individual beats.

After each beat of the ECG signal is obtained, Algorithm 1 is used to generate ECMs. Its input arguments are as follows.

- ECG: preprocessed and normalized ECG signal.
- R-peaks: extracted by Pan-Tompkins algorithm.

Algorithm 1 Generating ECM's Algorithm

```

1: procedure ECM-GENERATOR( $ECG, R_{peaks}, Ave_{dist},$   

    $init_{window}, bpf$ )
2:    $init_{seq} \leftarrow 0.2 \times Ave_{dist}$ 
3:    $end_{seq} \leftarrow 1.5 \times Ave_{dist}$ 
4:    $all\_segments \leftarrow []$   $\triangleright$  empty array for segments
5:   for each  $peak$  in  $R_{peaks}$  from  $init_{window}$  to  

    $init_{window} + (bpf - 1)$  do
6:      $segment \leftarrow ECG[peak - init_{seq} : peak +$   

    $end_{seq}]$ 
7:      $all\_segments \leftarrow all\_segments \text{ append } segment$ 
8:   Uniformize segments with zero padding, if needed.
9:   return  $ECM \leftarrow \text{Heatmap}(all\_segments)$ 

```

- Average distance: Mean interval between R-peaks in the user's ECG signal.
- initial window: shows the start point in the ECG to generate the next ECM.
- bpf: number of heartbeats per frame for generating ECM.

Each iteration selects a fixed bpf number to generate its respective ECM, ensuring that the R-peak is not at the beginning of the segment. Therefore, lines 2 and 3 of the algorithm address the signal selection where the R-peak falls almost in the middle of the segment. Specifically, we calculate the average distance between R-peaks in the entire ECG signal for each user. Then, we take 0.2 times this average distance before the R-peak of the segment and 1.5 times this distance after the R-peak. Subsequently, in each iteration, lines 5 and 6 of the algorithm extract multiple segments, which are then input into the rows of a matrix. The algorithm then transforms the matrix into an ECM using a heatmap procedure (line 9) [10]. Due to the varying lengths of segments, zero padding is applied to fill the matrix. The matrix has rows equal to the number of bpf-1 and columns equal to the largest segment. The output of Algorithm 1 is the corresponding ECM of the ECG.

Algorithm 2 Authentication Algorithm

```

1: procedure LEARNING-ALGORITHM( $ECG_{Datasets}$ )
2:   for dataset in  $ECG_{Datasets}$  do
3:     Apply Pan-Tompkins algorithm on the dataset
4:     Generate  $D_{ECMs}$  using Alg. 1
5:     for  $ecm_{no} = 5:70$  do
6:       Choose ECMs for all users from  $D_{ECMs}$ 
7:       Training(ECMs)
8:     Evaluate test results

```

2) *Proposed identification learning models:* After creating ECMs for each user, the next is to learn the identification models using ECMs. Algorithm 2 shows the user identification framework. The input argument is the ECG datasets. Preprocessing is applied to each ECG dataset (line 3) to extract R-peaks and other related properties. Subsequently, in line 4, ECMs of the corresponding users of each dataset are

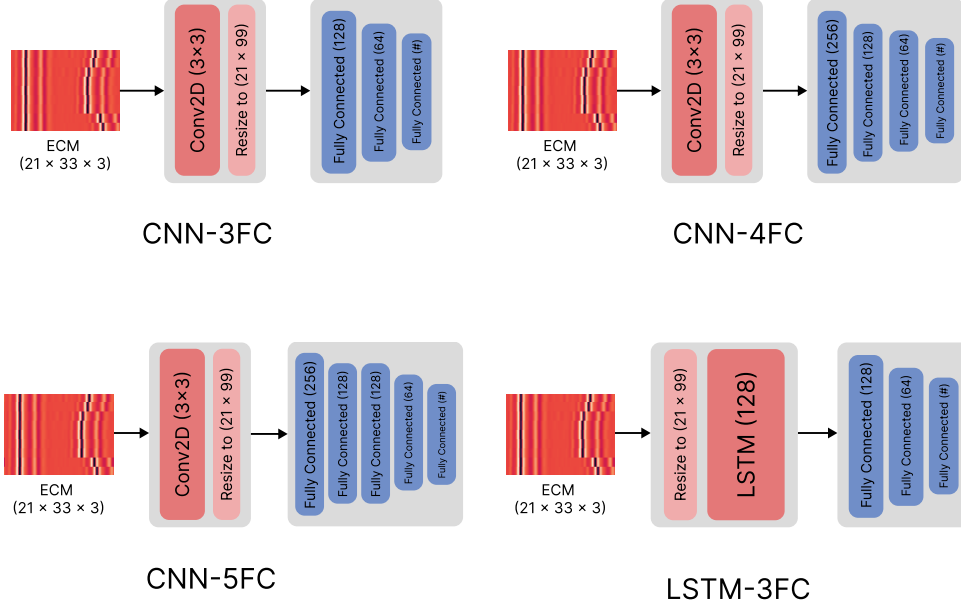


Fig. 1: Proposed models.

computed. To check the impact of the number of ECMs, each time we run the framework with a specific number of ECMs, $|ECMs| = ecm_{no}$ (lines 5 to 8). Notably, if a user did not have a sufficient number of ECMs, we used all the available ECMs for that user. This situation occurred in the PTBDB dataset due to the inadequate ECG signals for some users. Then, the identification model is learned using the generated ECMs for each specified number of ECMs. Finally, the results are evaluated on the test subset.

We employ four learning models within the framework outlined in Algorithm 2 to facilitate identification. Three of these models are CNN-based (Convolutional Neural Network), while the fourth is an LSTM (Long Short-Term Memory) model to assess the impact of recurrent models. Figure 1 shows the four models that have been proposed and used in this paper. Below, we provide a brief introduction to each of them.

- **CNN-3FC:** First, a convolutional layer is used, followed by a max pooling layer. Then, a dropout is applied. The output is then flattened and fed into three consecutive fully connected layers. Finally, a softmax layer is applied to identify the corresponding individual of the input ECM.
- **CNN-4FC:** All configurations of this model are the same as the previous model except that four fully connected layers are applied to the data.
- **CNN-5FC:** It utilizes five fully connected layers. This model uses five fully connected layers, while the other components remain the same as in CNN-3FC.
- **LSTM-3FC:** This model uses an LSTM learning model to take advantage of the correlated features of the data.

III. RESULTS

We ran the results on an Intel® Core™ i7-6700K CPU, 16 GB RAM system in a Python environment, using the TensorFlow library for learning models. The identification belongs to the classification paradigm; therefore, we use the cross-entropy loss function.

$$\text{Loss} = - \sum_{i=1}^M y_i \log(p_i) \quad (1)$$

Moreover, we used the following evaluation metrics to compare the proposed techniques fairly.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (2)$$

$$\text{AUC-ROC} = \sum_{i=1}^n TPR_i \Delta FPR_i, \quad (3)$$

$$\text{AUPR} = \sum_{i=1}^n \text{precision}_i \Delta \text{recall}_i. \quad (4)$$

As mentioned earlier, we have tested different scenarios to evaluate efficient identification. Therefore, different values of bps, ECMs, and learning models have been generated. Tables I to III compares the learning models for three different datasets. We have evaluated the results for two values of 10 and 20 ECMs. Table I shows the results for the MITDB. The CNN-FC4 has the best performance for $|ECM| = 10$. CNN-FC5 has the highest AUPR for $|ECM| = 20$. This parameter has a higher significance for imbalanced datasets and is more proper for evaluation.

Table II shows the results for the NSRDB. While the CNN-based methods have the highest AUPR and AUC-ROC, the

TABLE I: Results for 10 and 20 ECMs with 5 bpf for MITDB

ECM	Model	Loss	Acc	AUC	AUPR
10	Fuster [5]	1.9016	61.458	95.040	63.810
	CNN-FC3	1.2227	70.833	97.910	75.015
	CNN-FC4	1.0910	80.208	98.854	83.448
	CNN-FC5	1.7755	71.875	97.734	76.543
	LSTM-FC3	1.3967	72.917	98.060	79.862
20	Fuster [5]	0.8210	82.292	98.666	87.803
	CNN-FC3	0.7541	81.771	99.183	86.637
	CNN-FC4	1.2839	85.938	98.906	91.016
	CNN-FC5	0.9765	85.938	99.101	91.524
	LSTM-FC3	1.0860	77.604	98.883	86.052

TABLE II: Results for 10 and 20 ECMs with 5 bpf for NSRDB

ECM	Model	Loss	Acc	AUC	AUPR
10	Fuster [5]	1.1794	75	94.640	73.785
	CNN-3FC	0.8851	72.222	97.662	81.116
	CNN-4FC	0.5625	80.556	99.219	89.754
	CNN-5FC	0.6067	80.556	99.160	90.637
	LSTM-3FC	0.7589	83.333	98.420	83.271
20	Fuster [5]	0.2649	91.667	99.797	97.252
	CNN-3FC	0.4135	87.5	99.362	95.356
	CNN-4FC	0.2794	95.833	99.789	97.998
	CNN-5FC	0.1501	94.444	99.939	99.018
	LSTM-3FC	0.6247	81.944	98.799	87.311

LSTM method has the highest accuracy for $|ECM| = 10$. The CNN-based models are the winners for $|ECM| = 20$. Table III shows the results for the PTBDB. While the CNN-4FC and CNN-5FC have the highest AUC-ROC and AUPR, respectively, the LSTM method has the lowest loss value for $|ECM| = 10$. Like the previous datasets, the CNN-based models are the winners for $|ECM| = 20$.

TABLE III: Results for 10 and 20 ECMs with 5 bpf for PTBDB

ECM	Model	Loss	Acc	AUC	AUPR
10	Fuster [5]	1.1355	80.810	99.390	86.858
	CNN-3FC	0.9609	83.451	99.468	89.181
	CNN-4FC	1.0967	84.507	99.742	90.178
	CNN-5FC	1.0195	86.268	99.721	91.49
	LSTM-3FC	0.8570	82.923	99.554	91.02
20	Fuster [5]	0.7144	89.225	99.675	93.843
	CNN-3FC	0.5034	91.115	99.819	95.752
	CNN-4FC	0.4708	91.493	99.954	96.021
	CNN-5FC	0.4800	92.722	99.918	96.823
	LSTM-3FC	0.5597	88.658	99.817	94.738

We examined how changing the bpf number for each ECM affected the evaluation metrics. Figure 2 illustrates how accuracy changes with varying bpf values. The results for each bpf value are averaged across all ECM values. The accuracy of PTBDB remains consistent as bpf increases, indicating that

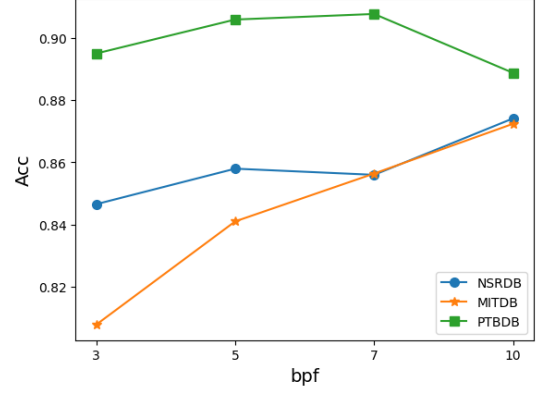


Fig. 2: Accuracy vs. bpf for LSTM-3FC

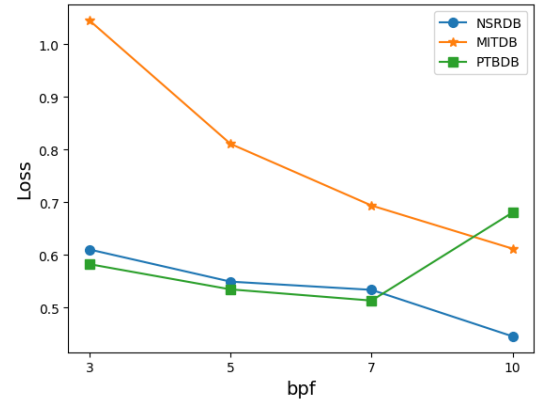


Fig. 3: Loss vs. bpf for LSTM-3FC

bpf does not impact its performance. In contrast, the accuracy for NSRDB slightly improves with higher bpf values. Finally, MITDB exhibits the most significant increase in accuracy as the bpf number increases.

Figure 3 illustrates the impact of bpf variation on the loss value. Although increasing the bpf on the PTBDB is negligible, it decreases the loss value of MITDB and NSRDB.

Figure 4 shows how the AUPR value changes as the

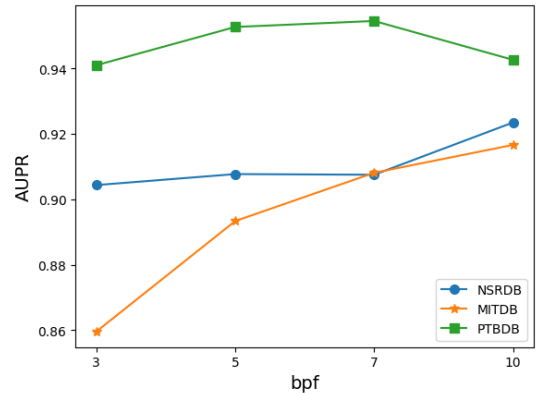


Fig. 4: AUPR vs. bpf for LSTM-3FC

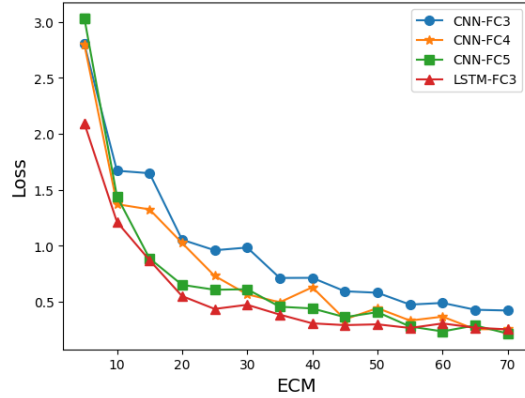


Fig. 5: Loss vs. ECM for LSTM-3FC

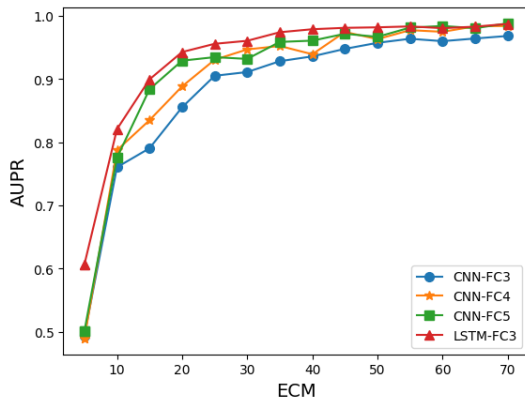


Fig. 6: AUPR vs. ECM

bpf value increases. The effect on PTBDB is minimal, but increasing the bpf value increases the AUPR of NSRDB and PTBDB. It's worth noting that the AUC follows the same pattern as the AUPR, so we will not include its figure.

Additionally, due to the limited amount of ECG signals available for some users in the PTBDB dataset, it was not possible to create enough ECMs with 10 bpf for some of the users. As a result, the evaluation metrics were not improved as expected.

As the last set of results, we evaluated the change in the number of ECMs on the model performance. Figure 5 illustrates the variation in the loss value against the ECM values. As the number of ECMs increases, the loss value of all models decreases.

The graph in Figure 6 illustrates how the AUPR value is affected by ECM. We observe a consistent improvement in performance as the number of ECMs increases. However, it becomes evident that all methods reach their peak performance for ECM numbers between 10 and 20. This pattern is also observed in other evaluation metrics, but their results are not included here.

IV. CONCLUSION

The study examines how deep learning models and the quantity of data inputted into them impact individual identification efficiency. We utilized three ECG datasets and transformed each user's ECG signal into ECMs. The findings indicate that, in most cases, CNN-based models outperform the LSTM model. Additionally, the increase in bpf generally has a negligible effect on the overall performance of the learning models.

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